MILITARY SPECIFICATION

AUGER, EARTH, SKIDMOUNTED; AND AUGER, EARTH, TRUCK MOUNTED, TURNTABLE BASE, POWER DRIVEN

This limited coordination Military Specification has been prepared by the Mobility Equipment Research and Development Command, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-A-516G. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers power driven earth augers equipped for mounting on and operating from standard military 5-ton trucks or commercial trucks. Each auger is equipped with pole-setting and wire-reeling equipment.

1.2 Classification. The augers shall be of the following types, as specified (see 6.2):

Type I - Skid mounted, gasoline engine driven.
Type II - Turntable base mounted, diesel engine driven.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.

FSC 3820

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FF-B-187	- Bearing, Roller, Tapered.
RR-W-410	- Wire Rope and Strand.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Low Leaded or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts for Internal Combustion Engines, Packaging of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Trucks; 5-Ton, 6 X 6, Military Design, M39 Series.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand-Operated, Level, Push, and Screw Type.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Tests.
MIL-STD-1474	- Noise Limits for Army Material.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35844	- Plug, Machine Thread; Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

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SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. Each earth auger shall be self-contained and shall consist essentially of an engine, a power transmission system and boring assembly. The augers shall be equipped with pole-setting equipment and a collapsible wire reel. Each auger with components shall not exceed the following overall dimensions:

Overall length	- 17 feet, 6 inches.
Overall width	- 92 inches for Type I, skid mounted auger. - 96 inches for Type II, turntable base mounted auger.
Height in travel position	- 81 inches.
Height in working position	- 18 feet.
Weight of complete auger	- Not to exceed payload rating or gross vehicle weight (GVW) rating of the specified truck.

3.1.1 Operational requirements and conditions. The augers shall perform all operational requirements specified herein at any temperature from plus 32° F to plus 125° F without exceeding any of the minimum or maximum operating temperature requirements specified herein.

3.2 First article (preproduction model). The contractor shall furnish one or more augers as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that the production methods will produce augers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

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3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more augers for testing as specified in 4.4.

3.3.1 Reliability assessment. When specified (see 6.2), the initial production test will be used to demonstrate the reliability of the auger. The demonstration test will be conducted in accordance with 4.6.2.3 and reliability assessed as specified in 4.6.2.11. Reliability will be expressed as Mean-Holes-Between-Failure (MHBF), and will be a point estimate of not less than 367 (MHBF).

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.14).

3.5 Safety. All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. All points requiring lubrication during operation shall have fittings located or guarded to as to be accessible without hazardous exposure to personnel performing these operations. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.6 Boring assembly.

3.6.1 Boring in unfrozen soil. When truck mounted, the augers shall:

(a) Bore holes 9 to 24 inches in diameter to a depth of 9 feet in all types of soil.
(b) Bore vertical holes 5 feet deep at an average rate of 85 seconds per hole when a 16-inch auger bit is used in a soil which is well-compacted, coarse-grained, well-graded, gravel-sand-clay mixture with an excellent binder and medium-to-high dry strength (see 6.4).
(c) Bore holes within the following limits:

    (1) From verical to 60 degrees away from truck.
    (2) From vertical to 10 degrees toward the truck.
    (3) From vertical to 30 degrees to the right and to the left of the truck.

(d) Have a down pressure of not less than 6,000 pounds.
(e) Have an up pressure of not less than 5,000 pounds.
(f) The Type II auger shall bore holes up to a diameter of 56 inches and to a depth of 10 feet in all types of soil.

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Depth of holes specified in (a), (b), and (f) above shall be measured from the ground level to the top of the loose dirt at the bottom of the hole.

3.6.2 Leveling mechanism. The boring assembly shall be equipped with a leveling device to permit the setting of the auger bar in a vertical position. The leveling mechanism shall be hydraulically operated. Leveling devices shall erect the auger bar to the operating position and lower it to the travel position.

3.6.3 Auger bar. The auger bar shall be housed in a steel tube covered at the top. The bar shall be straight within 1/8 inch for each 5 feet of length. A sheave shall be mounted at the top of the auger bar housing or boom for use in pole-setting.

3.6.4 Auger bits and teeth. Augers shall be of the double cutting type equipped with removable tungsten carbide teeth and pilot bits. One extra set of tungsten carbide teeth and pilot bits and two sets of dirt teeth and pilot bits shall be furnished with each size auger bit specified herein. Dirt teeth and pilot bits shall be alloy steel forgings, heat treated to a hardness of not less than 45 on the Rockwell C scale. Removal and replacement of teeth shall be accomplished using common handtools. Teeth shall be secured to the augers by a positive means which shall prevent detachment of the teeth from the augers while the auger bar is:

(a) Rotating clockwise.
(b) Rotating counterclockwise.
(c) Boring.
(d) Retracting.

One each of the following sizes of auger bits shall be furnished with each auger: 9-inch, 12-inch, 16-inch, and 24-inch. When specified (see 6.2), a 24-inch belling tool shall be furnished.

3.6.5 Controls. The auger shall be provided with the necessary means of control. Controls shall include but not be limited to an auger bar control, hand-feed lever, and emergency stop button. The auger bar control shall remain in a neutral position until pressure is applied to the lever. Positive disconnection of the hand-feed lever shall be provided by use of springs or other mechanical devices. An emergency stop button for stopping the engine shall be provided. All controls and the stop button shall be in accordance with SAE J833, J898, J899, and J925, shall be placed within reach of the operator, and shall permit a clear view of the boring operation. An identification plate shall be located on or adjacent to each control to indicate the function. The neutral and operating positions shall be shown by arrows and words. When the same lever or control serves more than one function, all functions shall be listed.

3.6.6 Transmission. The auger shall be equipped with an internal torque-coverter transmission. The torque-converter transmission shall have not less than four speeds forward and three speeds reverse. Engine and torque-conver-

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.ter transmission shall be matched for power as a unit. The torque-converter shall be provided with an oil cooler that shall prevent the temperature of the oil from exceeding 240° F. Augers equipped with torque-converter transmision shall be furnished with an oil temperature indicator and an oil pressure indicator. Indicators shall be located at the operator's station. Gear reductions for the four forward graduated speeds shall be such as to rotate the boring bar when it is above ground and rotating freely over a speed range from a minimum speed of not more than 40 revolutions per minute (rpm) in first gear to a maximum speed of not less than 120 rpm in highest gear.

3.6.7 Spoil throwoff. The auger bar speed shall be sufficient to throw spoil from the auger bits specified in 3.6.4, and deposit the spoil so that spoil will not fall back into the hole when the auger is digging maximum depth holes in the soil specified in 3.6.1 (b).

3.6.8 Bearings. Unless otherwise specified herein, all bearings shall be of the antifriction type and shall conform to FF-B-185, FF-B-187, or FF-B-171, as applicable. When specified (see 6.2), the contractor's standard commercial bearings may be used.

3.6.9 Hydraulic system. When a hydraulic system is used, it shall be protected against the entrance of dirt or other foreign materials. The hydraulic system shall be complete with all operating accessories, including a pump, hydraulic cylinders, valves, reservoir, pressure relief valve, and a full flow 25-micron oil filter accessible for cleaning. The pump shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. Hydraulic cylinders shall be of the double-acting type and shall be equipped with counterbalance or lock-type check valves. A wiper shall be provided on the cylinders adjacent to the piston rod seals to prevent dirt and abrasives from damaging the seal or entering the cylinder. Hydraulic cylinder piston rods shall be hard-chrome plated. Hydraulic valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. The reservoir shall be baffled and shall be constructed to withstand vibrations and shocks. The reservoir shall have sufficient capcity to allow maximum displacement of all cylinders, and aid heat dissipation. The return line shall terminate below the minimum oil level of the reservoir. The suction line shall be above the reservoir bottom and shall be protected with an accessible strainer. The reservoir shall be equipped with a removable cover plate for inspection and cleaning. Provisions shall be made for filling and draining the reservoir and for checking the quantity of fluid in the reservoir. The drainplug shall be of the magnetic type. The system shall have a shut off capability to prevent any excessive loss of oil while breaking lines or changing the pump.

3.7 Human engineering factors.

3.7.1 Characteristics. The characteristics of the auger shall provide for operation by personnel ranging from the small operator, clothed, through the large operator, arctic-clothed, in accordance with SAE J833.

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3.7.2 Noise limits. The noise of the auger (excluding horns and warning alarms) shall not exceed the limits specified in MIL-STD-1474, category C for steady-state noise when tested in accordance with 4.6.2.8.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.6.2.8.2 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.12), shall be provided if the sound level exceeds 85 dB(A) when tested as specified in accordance with 4.6.2.8.1. Hearing protectors shall not be supplied with the auger. Hazard signs, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED". Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material.

3.8 Pole-setting equipment and wire reel. The auger shall be equipped with a power winch, wire rope, sheaves, and wire reel.

3.8.1 Winch. The winch shall be complete with wire rope, sheaves, and other attachments necessary to convert the auger bar into a boom for setting power or telephone poles of the I-treated, 40-foot class. The winch drum shall have a capacity of not less than 150 feet of 7/16-inch-diameter wire rope. The line speed on a bare drum shall be not more than 15 feet per minute (fpm) at low speed nor less than 50 fpm at high speed. The single-line bare drum pull at maximum drum speed shall be not less than 3,000-pounds. The winch shall be provided with a brake that will control, stop, and hold a load of 4,500 pounds on the bare drum. Brake linings shall be riveted or bolted. The winch shaft shall be of sufficient length to support the wire reel on the outside of the skid frame.

3.8.1.1 Power takeoff. The winch shall be hydraulically powered by the auger engine to permit independent operation of either the auger or winch.

3.8.1.2 Wire rope. Wire rope shall conform to RR-W-410, type 1, class 2, 7/16-inch diameter, performed, uncoated, regular lay.

3.8.2 Wire reel. The wire reel shall be of the heavy-duty, collapsible type and shall rewind all sizes of spool wires. The reel shall be mounted on the winch shaft with a bayonet catch. Controls shall be provided to permit operation of the winch and reel exclusive of other components of the auger. The diameter of the expanded reel shall be not less than 28 inches.

3.9 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the auger shall be in accordance with SAE J551.

3.10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.10.1 Lubricants. Unless otherwise specified (see 3.10.6), the auger shall operate as specified herein when lubricated with military lubricants (see 6.5).

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3.10.2 Lubrication fittings. Unless otherwise specified herein, lubrication fittings shall conform to MIL-F-3541, types I, II, and III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibilty to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are used they shall have spring-loaded covers. Removable panels and plates equipped with hand-operable, quick-disconnect fasteners may be used to provide accessibility.

3.10.2.1 Optional fittings. When specified (see 6.2), the contractor's standard commercial lubrication fittings accessible to a standard commercial grease gun will be acceptable.

3.10.3 Pressure-release device. A suitable pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.10.4 Grease lubrication. Grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827 as applicable. The auger shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases may not be compatible with other greases. Each auger shall be tagged in a conspicuous place to indicate which military grease has been used.

3.10.5 Oil lubrication. Unless otherwise preserved for packaging or as specified herein, all components shall be filled to the operating level with grades of military oils designated for use in the temperature range of 10° F and above. Each auger shall be tagged in a conspicuous place to indicate the temperature range and grades of lubricants used.

3.10.6 Optional lubrication prior to delivery. When specified (see 6.2), the auger shall lubricated prior to delivery with those lubricants which are normally supplied or recommended for use with the auger when furnished commercially. The auger shall be conspicuously tagged to identify the lubricants used and their temperature range.

3.11 Lifting provisions. The auger shall be provided with permanently attached lifting provisions. The provisions shall enable the auger to be lifted in its normal traveling or operating position. The provisions shall be located so that not less than a 1-inch clearance is maintained between the slings and all exterior parts of the auger when being lifted. Each provision shall withstand without damage to any part of the auger not less than 2-1/2 times the normal load imposed on the provisions. Provisions for multiple slings shall be located so that the slings will converge not more than 24 feet above the lowest extremity of the load, and not less than 1-inch clearance will be maintained between the slings and the auger. Provisions shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the provision. Eye openings and clearance dimensions of the provision eyes shall conform to MIL-STD-209. Members to which provisions are fastened and the fasteners shall withstand all stresses imposed by the provision without damage

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or permanent deformation. When required, structural members shall be reinforced to meet the requirements specified herein. Lifting provisions may also be used as tiedown provisons when such provisions meet the requirements specified in 3.12.

3.12 Tiedown provisions. Augers shall be provided with provisions to permit tiedown or attachment to the floor, deck, or frame of the transportation medium. Each provision shall withstand without damage to the auger, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, and 1-1/2-G sideward, where G equals cargo weight of the auger. Provisions shall be rounded and beveled to prevent cutting of the tiedown material. Provisions may also be used as lifting provisions when such provisions meet the requirements specified in 3.11.

3.13 Identification marking. The auger shall be identified in accordance with MIL-STD-130. The marking shall be applied to the auger on identification plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. The location of all provisions and recommended lifting and tiedown arrangements shall be included on these plates.

3.13.1 Shipping data plates. A shipping data plate conforming to MIL-P-514, type II, composition C, of type I, grade A, class 1 material shall be provided. The plate shall also show the silhouette of the auger in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.13.2 Instruction plates. Instruction plates shall be furnished and shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, dimensions to suit the application. These plates shall indicate sequence for start up, shut down, and normal operation; and all inherent dangers which can stem from unsafe operation or an unsafe operating position. Where applicable the words "Warning" or "Caution" shall appear and shall be red. The shade of red shall conform to FED-STD-595 No. 31136.

3.14 Treatment and painting. Unless otherwise specified herein, the portions of the auger normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.14.1 Optional treatment and painting. When specified (see 6.2), treatment and painting shall be in accordance with the contractor's standard commercial practice.

3.15 Toolbox. The auger shall be provided with a toolbox made of metal not less than U.S. standard gage No. 14 (0.0747 inch thick) and of sufficient size to accommodate the operating and maintenance tools normally furnished with the auger. The toolbox shall have a continuous hinged lid and two fasteners to secure the lid. The toolbox shall be securely fastened to the auger in a protected position located so that the lid can be opened more than 90 degrees from the closed position. A hasp that will accommodate a padlock that has 3/8-inch-diameter shackle shall be provided for the toolbox lid. The toolbox shall shed water and have provision for draining.

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3.16 Maintainability. The auger shall operate as specified without maintenance other than the contractor's recommended schedule of adjustments as established by a maintenance schedule submitted by the contractor prior to test. All major assemblies shall be accessible for servicing, repair, and replacement without the removal of other major assemblies. Maximum use shall be made of interchangeable hardware and fasteners. All maintenance, assembly, or disassembly operations shall be accomplished with common handtools and those special tools furnished with the auger.

3.16.1 Filling, draining, and checking provisions. The hydraulic-system reservoir and transmission housings shall be equipped with dipsticks to determine the fluid level. Other enclosures such as gearcases and reservoirs which contain lubricant or hydraulic oil shall be equipped with dipsticks or sight gages not less than 1/2-inch pipe size to determine the fluid level. Each enclosure shall be equipped for filling and for draining. The drain outlet shall be fitted with a magnetic drainplug having either tapered pipe threads or shoulder and gasket, size conforming to MS35844 or MS49006. In lieu of a magnetic drainplug, a separate magnetic device may be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. Each drainplug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the auger is in a level position, drainage of fluids shall be accomplished without draining on or over any part of the auger. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid-level checking device, and the drainplug or valve shall be provided without removal or adjustment of accessories and parts, except plates equipped with hand-operable, quick-disconnect fasteners, or capscrews in application where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete draining of the system. The radiator drain shall be such that coolant will not drain onto any part of the auger or carrier. An extension hose is acceptable to facilitate radiator drainage.

3.17 Type I, skid mounted. The type I, skid mounted auger shall be powered by a gasoline engine and shall be designed for military truck mounting (see 3.17.1) or commercial truck mounting (see 3.17.4), as specified (see 6.2).

3.17.1 Military truck mounting. When specified (see 3.17), the auger shall be designed for mounting on a 5-ton, military design truck conforming to MIL-T-740 or equal. The auger shall be furnished truck mounted or unmounted as specified (see 6.2). When furnished unmounted, the auger shall be furnished with all accessories required to mount the auger on the military truck. When furnished truck mounted the Government will furnish the military truck for mounting the auger; and when furnished unmounted the Government will loan a military truck to facilitate testing the auger (see 6.6 and 6.7).

3.17.2 Mounting accessories. When furnished truck mounted, the auger shall be mounted on the truck using six sets of twin bolts with clips to clamp the skid frame of the auger to the frame of the truck. Two of the six sets of twin bolts

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shall be provided with clips for attaching stabilizing jacks (see 3.17.3) under the rear corners of the truck body. No welding or cutting of the truck frame shall be required to install and remove the auger from the truck. When the auger is furnished unmounted a complete set of mounting accessories and two jacks as specified in 3.17.3 shall be furnished with the auger.

3.17.3 Stabilizing jacks. Two telescoping-type stabilizing jacks shall be furnished to support the rear of the truck or truck chassis-cab during drilling operations. The stabilizing jacks shall attach to the frame of the truck with bolts and clips provided as mounting accessories and shall have a working range of 14 inches above level ground when retracted to 10 inches below level ground when extended.

3.17.4 Commercial truck mounting. When specified (see 3.17), the type I auger shall be furnished mounted on a commercial truck. The truck shall be of the type, size, capacity, and have accessories, as specified (see 6.2).

3.17.5 Gasoline engine. The gasoline engine shall be a heavy duty commercial industrial type, capable of operating on fuel conforming to VV-G-1690, and lube oil conforming to MIL-L-2104. The average oil consumption during preproduction and initial production testing shall not exceed 0.0035 pounds per brake horsepower hour. Full-flow type oil filters shall be utilized in the engine lubricating system. The temperature of the oil in the engine oil gallery shall not exceed 230° F when tested as specified. When an oil cooler is provided, the temperature of the oil in the engine sump, or entering the cooler, shall not exceed 250° F when tested as specified. The temperature of the coolant in the top tank of the radiator shall not exceed 210° F when tested as specified. The engine shall be the same engine that is being used and has been used by the contractor in his commercial auger of the same size and capacity for a period of not less than one year. The engine shall be equipped with the following:

(a) A fuel tank of sufficient capacity for 8 hours of continuous rated operation.
(b) A 12- or 24-volt, negative ground, electrical starting system including a battery charging system (generator or alternator) of not less than 18-ampere capacity. Batteries shall conform to MS35000.
(c) A steel housing, complete with side panels.
(d) Instrument panel with all instruments specified below located within full view of the operator during digging operations.

(1) Engine lubricating oil temperature gage.
(2) Engine coolant temperature gage.
(3) Tachometer in accordance with SAE J678.
(4) Ammeter, battery-generator.
(5) Hourmeter, with time totalizing mechanism of not less than 9,999 hours in accordance with SAE J678.
(6) Fuel quantity indicator.
(7) Safety control to shut down engine when coolant temperature reaches 220° F.

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(8) Safety control to shut the engine down when engine oil pressure drops below 15 psi.
(9) Safety control to shut the engine down when the engine speed exceeds engine manufacturer recommended maximum speed.

(e) A weatherproof battery box painted with acid resistant paint.

3.17.5.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the auger under any of the operating conditions specified herein, will not require horsepower in excess of the continuous rating of the engine as published by the engine manufacturer, or 85 percent of the engine manufacturer's published maximum horsepower rating, at the applied speed; whichever is less.

3.17.6 Auger bar. The type I auger bar shall be lowered and raised by a hydraulically controlled mechanism. The auger bar shall be provided with an hydraulic or mechanical crowd mechanism. The rotary drive shall be either mechanical or hydraulic. The auger bar shall be hot rolled AISI-4145 fine grain steel, quenched and tempered to 302-341 Brinell hardness and thermally stress relieved within a temperature range of 950° to 1,000° F.

3.17.7 Skid frame. The skid frame shall mount directly over a standard SAE J691, 34-inch truck frame and shall furnish support for the auger. The skid frame shall be provided with towing attachments on each end and a mounting for the power winch and wire reel. The skid frame with auger shall not snub, break, or be permanently deformed when towed over rough terrain at a speed of 5 mph.

3.18 Type II, turntable base mounted. The type II turntable base mounted auger shall be powered by a diesel engine and shall be furnished mounted on a commercial truck. The truck shall be of the type, size, capacity, and have *accessories, as specified (see 6.2).*

3.18.1 Diesel engine. The diesel engine shall be a heavy duty commercial industrial type, capable of operating on diesel fuel conforming to VV-F-800 and lube oil conforming to MIL-L-2104. The average oil consumption during preproduction and IPT testing shall not exceed 0.0035 pounds per brake horsepower hour. Full-flow type oil filters shall be utilized in the engine lubricating system. The temperature of the oil in the engine oil gallery shall not exceed 230° F when tested as specified herein. When an oil cooler is provided, the temperature of the oil in the engine sump, or entering the cooler, shall not exceed 250° F when tested as specified herein. The temperature of the coolant in the top tank of the radiator shall not exceed 210° F when tested as specified herein. The engine shall be the same engine that is being used and has been used by the contractor in his commercial auger of the same size and capacity for a period of not less than one year. The engine shall be equipped with the following:

(a) Fuel tank of sufficient capacity for 8 hours normal operation.
(b) A two-stage dry type intake air cleaner with service indicator.
(c) Exhaust muffler and stack with rain cap or 90 degree bend in the outlet.

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(d) A 12-volt negative ground direct current cranking system with battery. The battery shall have a capacity of not less than 150 ampere hours at a 20-hour rate.
(e) A battery charging generator or alternator with a rating of not less than 30 amperes.
(f) Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.
(g) Hood and side doors.
(h) A mechanical liquid quantity fuel indicator located at the tank.
(i) Hour meter.
(j) An instrument and control panel located at the operator's station and equipped with the following indicators and controls:

(1) Water cooling temperature indicator.
(2) Ammeter.
(3) Lubricating oil pressure indicator.
(4) Tachometer.
(5) Manual speed control.
(6) Emergency shutdown device.

(k) A weatherproof battery box painted with acid-resistant paint.

3.18.1.1 Power and speed rating. The power and speed rating of the diesel engine shall be such that operation of the auger under any of the operating conditions specified herein, will not require horsepower in excess of the continuous rating of the engine as published by the engine manufacturer, or 85 percent of the engine manufacturer's published maximum horsepower rating, at the applied speed; whichever is less.

3.18.2 Auger bar. The type II auger bar shall be provided with an hydraulically actuated crowd mechanism. Bar material shall be an alloy steel as furnished commercially by the contractor. The auger bar housing or boom shall be hydraulically extendible. Extension travel shall be not less than 5 feet. When fully extended, the distance from the ground to the top of the boom shall be not less than 27 feet with a lifting capacity of not less than 5,000 pounds. The boom shall be equipped with a hydraulically operated, circumferential type, pole grabber. The pole grabber shall maintain a constant clamping force at any angle of operation. Boom travel and pole grabber shall be controlled by operator at the operator's station.

3.18.3 Turntable. The turntable subframe shall provide for rotating the auger not less than 90 degrees to the right and left of the truck's longitudinal axis. Fore and aft travel of the turntable frame shall be not less than 22 inches. Rotation of the turntable and fore and aft travel of the frame shall be hydraulically operated and shall be controlled by the operator at the operator's station. The turntable base shall be installed on the truck by the use of removable locating mechanisms that fit into the stake pockets. The base shall be constructed of steel plate, reinforced to withstand the maximum load and stress imposed. Provisions for lubricating the center pin shall be provided.

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3.18.4 Stabilizing jacks. The turntable base shall be equipped with not less than two hydraulically operated stabilizing jacks. Two jacks shall be located at the rear corners of the turntable base and may be installed on outriggers. The lowest extermity of the jacks, when fully retracted, shall not extend below the rear axle of the truck on which the auger is mounted. Operation of the jack shall be controlled by the operator at the operator's station.

3.19 Workmanship. The auger shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the auger. Metal used in the fabrication of the auger shall be free of kinks and sharp bends. Shearing and clipping shall be done neatly and accurately. Flame cutting of steel may be employed. Burned surfaces of flame-cut steel shall be ground or machined to remove all ash, burrs, and rough edges. All bends shall be made with metal dies or fixtures to insure uniformity in size and shape. Rivets shall fill the holes completely. Rivet heads shall be full, neatly made in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.19.1 Welders and welding.

3.19.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder is qualified in accordance with either of the following listed codes for the method, procedure, materials and joint configurations to be welded and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
Welding Qualifications of the ASME, Section IX.

Welders who only make horizontal welds need not qualify for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that manual automatic welder equipment operators are capable of producing welds that conform to the code.

3.19.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.19.2 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extreneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproducton inspection.

4.3.1 Engine performance. Prior to examination and test of the auger, the engine which will be installed in the preproduction model shall be tested in accordance with test method 2000 of MIL-STD-1400. Failure to meet any of the operational and performance requirements specified in 3.17.5 or 3.18.1 shall constitute failure of this test.

4.3.2 Examination. The auger shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The auger shall be tested as specified in 4.6.2.1 through 4.6.2.9. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production augers will be selected at random by the be Government from augers being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.9 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production auger shall not exclude the remaining augers from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.4.1 Inspection failure. Failure of an initial production auger to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production auger and shall be cause for refusal by the Government to continue acceptance of augers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on augers previously accepted and produced under the contract. Any deficiencies found as a result of initial production inspection will be considered prima facie evidence that all augers accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each auger shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each auger shall be tested as specified in 4.6.2.10. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The auger shall be examined as specified herein, for the following defects:

101. Dimensions or weight not as specified.
102. Fasteners not as specified.
103. Parts or components missing, do not function or not as specified.
104. Safety provisions not as specified.
105. Material not as specified.
106. Lubrication devices not as specified.
107. Workmanship not as specified.
108. Treatment and painting not as specified.
109. Identification or other marking, operating and safety instructions, or manuals missing.
110. Auger bar not straight within 1/8 inch for each 5 feet of length.
111. Maintainability not as specified.

4.6.2 Tests

4.6.2.1 Test conditions (operating). Prior to test, the auger shall be lubricated with oils and greases as specified in 3.10.1 or 3.10.6 as applicable. Oils shall be those designated for use in the ambient temperature at which the test is being performed. All tests except 4.6.2.7 and 4.6.2.9 shall be conducted with the auger mounted on the applicable truck specified in 3.17.1, 3.17.4, or 3.18. All drilling and measurements for test purposes shall be done with the truck on a level surface.

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4.6.2.2 Transmission speeds. Operate the auger with the auger bit above the ground and with the engine at full-load governed speed. All forward speeds and the reverse speeds shall be used. Measure the revolutions per minute of the boring bar while operating in first gear and highest gear. Rotation of the boring bar at a speed greater than 40 rpm in first gear or less than 120 rpm in highest gear shall constitute failure of this test.

4.6.2.3 Performance. The auger shall bore holes in soil in accordance with table I. All vertical holes shall be bored to a depth as specified in table I. All angle holes shall be bored to the depth that results when the auger bar is extended the same distance that it was extended to drill the vertical holes.

TABLE I. Boring tests.

Test No.	Auger position	Type I auger			Type II auger		
		Bit size inches	Hole depth feet	No. holes bored	Bit size inches	Hole depth feet	No. holes bored
1	Vertical	24	9	200	56	10	10
2	10 degrees from vertical toward truck	24	-	200	24	--	10
3	60 degrees from vertical away from truck	24	-	200	24	--	10
4	30 degrees to the left	24	-	200	24	--	10
5	30 degrees to the right	24	-	200	24	--	10
6	Same as for tests 1 through 5	16	-	15	16	--	10
7	Same as for tests 1 through 5	12	-	5	12	--	10
8	Same as for tests 1 through 5	9	-	5	9	--	10

Inability to bore holes to the depth and diameter as specified in table I; inability to bore holes at angles of 10 degrees, 30 degrees, or 60 degrees as specified above; inabilty to bore vertical holes with the 16-inch bit at an average rate of 5 feet in 85 seconds; inability to clear spoil from the auger bits and remove the spoil so that it does not fall back into the hole; evidence of breakage, permanent deformation, leakage of hydraulic oil, or evidence that the replaceable points or cutting edges of the bits are not highly-wear resistant shall constitute failure of this test.

4.6.2.4 Down pressure. Operate the auger to apply a downward force of not less than 6,000 pounds. This test shall be repeated 50 times. Inability to apply a downward force of not less than 6,000 pounds each time or evidence of any structural failure of the auger assembly (including skids) shall constitute failure of this test.

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4.6.2.5 Pull. With the auger bit anchored to the ground, place the boring bar control in the raised position and operate to exert not less than 5,000 pounds raising effort at the boring bar. Inability to exert not less than 5,000 pounds raising effort or evidence of any structural failure of the auger assembly (including skids) shall constitute failure of this test.

4.6.2.6 Pole-setting and winch operation.

4.6.2.6.1 Pole-setting. Set a power or telephone pole of the I-treated, 45-foot class in a vertical hole using the pole grabber and auger bar housing or derrick as a boom. Inability to set the pole in a vertical position, or damage or permanent deformation to the auger or pole-setting equipment shall constitute failure of this test.

4.6.2.6.2 Winch operation. Operate the winch drum for 15 minutes with the engine running at approximately one-third speed and 15 minutes with the engine running at maximum speed. Then test the brake by the lifting, lowering, stopping, and holding a load of 4,500 pounds on the bare drum. Inability of the drum to turn, excessive noise or grinding, inability of the drum to operate independently of the auger, or failure of the brake to stop and hold a load of 4,500 pounds shall constitute failure of this test.

4.6.2.6.3 Drum capacity. Wind a 7/16-inch diameter wire rope on the drum until the drum is filled to capacity. Measure the length of the wire rope that is wound on the drum. A length of less than 150 feet shall constitute failure of this test.

4.6.2.6.4 Single-line pull. With the winch drum bare, raise a 3,000-pound load, using a single line. A line speed of less than 50 fpm or a line pull of less than 3,000 pounds shall constitute failure of this test.

4.6.2.7 Electromagnetic interference. The preproduction model shall be tested in accordance with SAE J551. The contractor shall furnish to the contracting officer the test plan and test report. Disapproval of the test plan or test report shall constitute failure of this test.

4.6.2.8 Sound level measurement. Test the auger as follows and nonconformance to 3.7.2 shall constitute failure of this test.

4.6.2.8.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state at operating conditions while boring as specified in 4.6.2.3.

4.6.2.8.2 Exterior spectator. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise.

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4.6.2.9 Tiedown and lifting provisions. The static load as specified in 3.11 for lifting provisions and in 3.12 for tiedown provisions shall be applied to the provision when mounted on its structural member. Weld failure or permanent deformation of the provision or the structrual member shall constitute failure of this test.

4.6.2.10 Operation. Operate each auger for a period of 1/2-hour without boring holes. Then operate the auger with bit above ground and engine at full load governed speed using all forward and reverse speeds. Operate the auger in the vertical, 10 degrees from vertical toward the truck, 60 degrees from vertical away from the truck and 30 degrees to the right and left of the truck. Operate the auger bar control, hand feed lever, and emergency stop. Rotate the turntable on type II augers not less than 90 degrees to the right and left of the longitudinal axis of the truck. Forward and reverse travel of the turntable frame on type II augers shall be not less than 22 inches. Operate the stabilizing jacks and the power-take-off. Evidence of breakage, permanent deformation, or any operating difficulty shall constitute failure of this test.

4.6.2.11 Reliabilty assessment. Reliability will be assessed from data derived during the test of 4.6.2.3. Reliability will be a point estimate and will be calculated as follows: Total holes bored divided by total number of failures. A failure will be defined as a random malfunction (dependent malfunctions are excluded) which causes degradation below operating condition required to continue the test and which cannot be remedied within 1 hour using resources available to the parent using unit or organization when deployed.

4.7 Inspection comparision. The Government may select augers at any time during the contract production period and subject these augers to the examination specified in 4.6.1 and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production augers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests, contained in this specification as part of the inspection comparison failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison auger to meet any requirements specified herein during and as a result of of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison augers and shall be cause for refusal by the Government to continue acceptance of production augers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on augers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all augers accepted prior to the completion of inspection comparison are similarly deficient unless evidence

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to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. When specified (see 6.2), the preproduction pack of the crated auger for level A or B shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or braking of holddowns, ties, stays, blocking or bracing, or any visual damage to the auger shall constitute failure of the preproducton pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Materials, methods, or containers not as specified.	Each incorrect material, method or container shall constitute one defect.
113.	Unpainted exterior metal surfaces not coated with preservative as specified for level A or B.	
114.	Engine and engine accessories not preserved in accordance with the referenced document.	
115.	Gears not preserved as specified for level A or B.	
116.	Clutch not preserved as specified for level A or B.	
117.	Exposed chains not preserved as specified for level A or B.	
118.	Hydraulic system not preserved and tagged as specified for level A or B.	
119.	Wire rope not preserved as specified for Level A.	
120.	Wire rope not wound on the drum and secured as specified for level A or B.	
121.	Repair parts, technical publications when applicable, or maintenance tools not preserved in accordance with the referenced documents for level A or B.	

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122. Consolidation not as specified for level A or B.
123. Packing not as specified for Level A or B.
124. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test within the time frame specified, to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The preproduction pack may be accomplished utilizing either the preproduction model auger or a production auger, If the preproduction model auger is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model auger and production augers.

5.2 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss. Removed bolts, nuts, pins, and washers shall be reinstalled in one of the mating parts and secured to prevent their loss. Removed keys shall be secured in the keyway.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of components of the auger such as levers, adjusting linkages, stabilizing jacks, auger bar housing, auger bits and teeth, mounting accessories, bolts, washers, nuts, and similar components not specified herein requiring a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I.

5.2.1.5 Gears.

5.2.1.5.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative.

5.2.1.5.2 Enclosed gears. Gear housings shall be filled to the operating level with the operational lubricant and gears actuated to assure coating all interior parts and surfaces with the lubricant.

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5.2.1.6 Clutch. With the cover plates removed and the clutch engaged, all interior components of the clutch shall be sprayed with primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutch shall be blocked in a partially disengaged position to prevent contact between the disk facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged. The clutch shall not be operated after application of preservatives.

5.2.1.7 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or type P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Sprockets shall be coated with type P-1 preservative.

5.2.1.8 Hydraulic system. The hydraulic fluid supply reservoir shall be filled to the operating level with the operational fluid. The pistons shall be retracted as fas as practicable into the cylinders and secured. Where the pistons cannot be fully retracted, exposed portions of the piston rods shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, or type I, grade C, class 1, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV. Where the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified for the piston rods. Hose shall not be disconnected. Any other exterior metal surfaces of components of the system requiring the application of a contact preservative shall be coated with type P-1 preservative. A tag conforming to UU-T-81, type A, shall be attached to the control lever indicating "The hydraulic supply tank is filled with the fluid required for operation." The tag and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.9 Winch. Components of the winch shall be preserved as specified herein for parts and assemblies having similar characteristics.

5.2.1.10 Wire rope. The wire rope shall be removed from the winch and coated with type P-1 preservative applied by brushing or by immersion. The rope shall be rewound on the winch drum and secured to prevent unwinding.

5.2.1.11 Technical publications. Technical publications, when applicable, shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.12 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the engine repair parts shall be preserved in accordance with level A of MIL-R-169.

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5.2.1.13 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40 and placed in the toolbox.

5.2.1.14 Consolidation. Loose components such as auger bits, mounting accessories, repair parts, and technical publication shall be packed in close-fitting boxes conforming to PPP-B-621, class 1, style optional, or PPP-B-601, domestic type, style optional. Cushioning and blocking to prevent movement and damage to the contents shall be provided, in accordance with MIL-STD-1186. Strapping shall not be required.

5.2.2 Level B. Unless otherwise specified (see 6.2 and 6.9), the auger components shall be preserved as specified in 5.2.1 for level A, with the following exceptions.

5.2.2.1 Clutch. Spring-loaded-type clutches shall be blocked in a partially disengaged position to prevent contact between the disk facings and pressure plates. Snap-over-center and toggle-in type clutches shall be completely disengaged. Preservation of the clutch components shall not be required.

5.2.2.2 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or type P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings.

5.2.2.3 Hydraulic system. The hydraulic system shall be preserved, wrapped, and tagged as specified in 5.2.1.8, except waterproofing of the tag shall not be required.

5.2.2.4 Wire rope. The wire rope shall be wound on the winch drum and secured to prevent unwinding. Preservation of the wire rope shall be as specified in 5.2.1.10.

5.2.3 Level C.

5.2.3.1 Alternate preservation. Unless otherwise specified herein, the auger, accessories, repair parts, technical publications, and maintenance tools shall be preserved and to afford protection against deterioration and damge from the contractor to the initial destination. The contractor's standard practice may be used when it fulfills these requirements. When specified (see 6.2), the following alternate preservation shall be performed.

5.2.3.1.1 Engine. The engine and liquid cooling system shall be preserved in accordance with the level C requirements of MIL-E-10062.

5.2.3.1.2 Repair parts, technical publications, and maintenance tools. Repair parts and technical publications shall be preserved as specified for level A. Maintenance tools do not require preservation but shall be placed in the toolbox.

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5.2.3.1.3 Other components. Preservation of other components not specifically mentioned herein shall be limited to that protection deemed necessary by the contractor to provide protection against deterioration and damage from the contractor to the initial destination. The contractor's standard practice is acceptable when it fulfills these requirements.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Each complete auger with the consolidated items shall be packed in a crate conforming to MIL-C-104, type I or II, class 1 or 2, style A. The consolidation containers shall be secured to the auger or to the crate base. The components shall be arranged within the crate in a manner to provide maximum compactness, low center of gravity, and in a manner that void spaces are held to a minimum. The contents shall be blocked, braced, and anchored in accordance with MIL-STD-1186 and the appendix to the crate specification.

5.3.2 Level B. Each complete auger with the consolidated items shall be packed in a crate conforming to MIL-C-3774, type I, style A. The consolidated containers shall be secured to the auger or to the crate base. The components shall be arranged within the crate in a manner to provide maximum compactness, low center of gravity, and in a manner that void spaces are held to a minimum. The contents shall be blocked, braced, anchored, and protected with a shroud in accordance with MIL-STD-1186.

5.3.3 Level C. Each complete auger shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules. Loose or unit packed components shall be secured to the auger or to the base of the container in a manner so as not to increase cubage.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

5.5 Depreservation guide. Two depreservation guides, DA Form 2258, shall be completed for each item in accordance with instructions on the form. One copy of the completed form shall be placed in a waterproof envelope marked "Depreservation Guide" and the envelope shall be attached to the operator controls. The other copy shall be placed in the package of technical manuals and the package marked to show that the depreservation guide is inside.

6. NOTES

6.1 Intended use. The earth auger is intended for use in the rapid boring of holes in earth, and when equipped with pole-setting and wire-reeling equipment may be used as a maintenance unit on communication lines and power lines.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of auger required (see 1.2).
(c) The time frame required for submission of preproduction model, and number of augers required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of augers to be furnished (see 3.3).
(f) When reliability assessment is required (see 3.3.1).
(g) When a belling tool is required (see 3.6.4).
(h) When standard commercial bearings are acceptable (see 3.6.8).
(i) When electromagnetic interference other than as specified is required (see 3.9).
(j) When optional lubrication fittings are permitted (see 3.10.2.1).
(k) When optional lubrication is required (see 3.10.6).
(l) Color of paint required (see 3.14).
(m) When treatment and painting in accordance with the contractor's standard commercial practice is required (see 3.14.1).
(n) Whether type I auger shall be designed for military truck mounting or commercial truck mounting (see 3.17).
(o) Whether the type I auger designed for military truck mounting shall be furnished truck-mounted or unmounted (see 3.17.1).
(p) Type, size, and capacity of truck and accessories to be furnished with the truck when the type I commercial truck mounted auger or the type II auger is being furnished (see 3.17.4 and 3.18).
(q) When testing of preproduction pack is required (see 4.8.1.2).
(r) Level of preservation and level of packing required (see 5.2 and 5.3). Level B preservation is intended to provide economical but limited protection and should be specified only when it is determined that the auger will be held in covered storage (see 6.9).
(s) When preservation of engine repair parts in accordance with MIL-R-196 is required (see 5.2.1.12).
(t) When level B preservation of auger components other than as specified is required (see 5.2.2).
(u) When alternate preservation for level C is required (see 5.2.3.1).

6.3 Preproduction model. Any changes or deviations of production augers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor's obligation to furnish augers conforming to this specification.

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6.4 Soil classification. The soil specified herein is of the type designated as GC in table I of SAE J106 (see 3.6.1 (b)).

6.5 Lubricants. When applicable, the contracting officer should furnish a list of military lubricants applicable to the auger covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification list C9100-IL, for FSC group 91.

6.6 Government-furnished property. The contracting officer should arrange to furnish the 5-ton, military design truck conforming to MIL-T-740, when required (see 3.17.1 and 6.2 (n)).

6.7 Government-loaned property. The contracting officer should arrange to loan the 5-ton, military design truck conforming to MIL-T-740, when required (see 3.17.1 and 6.2 (o)).

6.8 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the augers in accordance with this specification.

6.9 Level B preservation. Level B preservation (see 5.2.2) has been developed for use by the Army and Air Force for the specific conditions specified in 6.2 (r). For Navy procurement, details of level B preservation when required, will be developed at time of procurement when all conditions are known.

6.10 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each auger.

6.11 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of the augers.

6.12 Noise limits. When the noise level of the auger exceeds 85 db(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (Manuals Paragraph) will include the requirement for hearing protection, the noise level of the auger, the distance from the auger at which 85 db(A) will always be met, and what operational situation or system configuration will reduce the level to 85 db(A).

6.13 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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